Light

GCSE Computer Science — Networks Basics (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 56 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A school has 200 computers that need to print documents.
What is one benefit of connecting these computers to a network with shared printers?

- ○ **A.** Multiple computers can use the same printer, reducing hardware costs
- ○ **B.** Each computer requires its own dedicated printer
- ○ **C.** Documents can only be printed when the network is disconnected
- ○ **D.** Printers will work faster when not connected to a network

Q2 (1 mark)

A business has computers connected together within a single office building.
What type of network is this?

- ○ **A.** A Wide Area Network (WAN)
- ○ **B.** A Local Area Network (LAN)
- ○ **C.** The internet
- ○ **D.** A Personal Area Network (PAN)

Q3 (1 mark)

A company has offices in London, New York and Tokyo. All offices are connected so employees can share files.
What type of network connects these offices?

- ○ **A.** A Local Area Network (LAN)
- ○ **B.** A bus network
- ○ **C.** A star network
- ○ **D.** A Wide Area Network (WAN)

Q4 (1 mark)

Every device connected to the internet needs a unique identifier to send and receive data.
What is this unique identifier called?

- ○ **A.** A MAC address
- ○ **B.** An IP address
- ○ **C.** A URL

- ○ **D.** A domain name

Q5 (1 mark)

Data sent across the internet passes through multiple devices that direct it towards its destination.
What network device performs this function?

- ○ **A.** A router
- ○ **B.** A printer
- ○ **C.** A monitor
- ○ **D.** A keyboard

Q6 (1 mark)

A network administrator is choosing between wired and wireless connections for desktop computers that will not be moved.
Which is an advantage of using wired connections?

- ○ **A.** Wired connections allow devices to move freely around the building
- ○ **B.** Wired connections typically provide faster and more stable data transfer
- ○ **C.** Wired connections do not require any cables
- ○ **D.** Wired connections have a longer range than wireless

Q7 (1 mark)

A coffee shop offers free internet access for customers using laptops and smartphones.
Which type of connectivity would be most suitable for this environment?

- ○ **A.** Wired Ethernet connections at each table
- ○ **B.** Fibre optic cables to each device
- ○ **C.** Dial-up modem connections
- ○ **D.** Wireless (Wi-Fi) connectivity

Q8 (1 mark)

A student reads that their home broadband has a bandwidth of 100 Mbps.
What does bandwidth measure?

- ○ **A.** The physical length of the network cable
- ○ **B.** The number of devices connected to the network
- ○ **C.** The age of the network equipment
- ○ **D.** The maximum amount of data that can be transmitted per second

Q9 (1 mark)

A user types a web address into their browser and presses Enter.
Which protocol is used to request and receive the web page?

- ○ **A.** FTP
- ○ **B.** SMTP
- ○ **C.** HTTP

- ○ **D.** POP3

Q10 (2 marks)

In a school network, all computers connect to a central switch.
What network topology does this describe?

- ○ **A.** Bus topology
- ○ **B.** Ring topology
- ○ **C.** Star topology
- ○ **D.** Mesh topology

Q11 (2 marks)

A student makes the following statements about LANs and WANs:
Statement 1: A LAN typically covers a small geographical area such as a building
Statement 2: A WAN is usually owned and maintained by a single organisation
Statement 3: The internet is an example of a WAN
Which statements are correct?

- ○ **A.** Statements 1 and 2 only
- ○ **B.** Statements 1 and 3 only
- ○ **C.** Statements 2 and 3 only
- ○ **D.** All three statements are correct

Q12 (2 marks)

A file has a size of 240 megabits. The network connection has a speed of 60 Mbps.
How long will it take to transfer this file?

- ○ **A.** 14,400 seconds
- ○ **B.** 0.25 seconds
- ○ **C.** 240 seconds
- ○ **D.** 4 seconds

Q13 (2 marks)

A gamer complains that there is a delay between pressing a button and seeing the action happen on screen during online play.
Which network characteristic is causing this problem?

- ○ **A.** High bandwidth
- ○ **B.** High latency
- ○ **C.** Strong signal strength
- ○ **D.** Large file size

Q14 (2 marks)

When shopping online, a user notices the web address begins with 'https://' instead of 'http://'.
What does the 's' in HTTPS indicate?

- ○ **A.** The connection is encrypted for security
- ○ **B.** The website loads at a slower speed
- ○ **C.** The website is stored on a server

- ○ **D.** The website was created recently

Q15 (2 marks)

A web developer needs to upload website files from their computer to a web server.
Which protocol is specifically designed for this task?

- ○ **A.** HTTP
- ○ **B.** SMTP
- ○ **C.** FTP
- ○ **D.** IMAP

Q16 (2 marks)

A user clicks 'Send' on an email they have written.
Which protocol transfers this email to the recipient's mail server?

- ○ **A.** POP3
- ○ **B.** IMAP
- ○ **C.** SMTP
- ○ **D.** HTTP

Q17 (2 marks)

The TCP/IP model organises network communication into layers.
Which layer is responsible for routing data packets across different networks?

- ○ **A.** Internet layer
- ○ **B.** Application layer
- ○ **C.** Transport layer
- ○ **D.** Link layer

Q18 (2 marks)

A network uses a single main cable with all devices connected to it. If the main cable fails, the entire network stops working.
What topology is this network using?

- ○ **A.** Star topology
- ○ **B.** Mesh topology
- ○ **C.** Bus topology
- ○ **D.** Ring topology

Q19 (2 marks)

A network is designed so that if any single connection fails, data can still reach its destination through alternative routes.
Which topology provides this resilience?

- ○ **A.** Mesh topology
- ○ **B.** Bus topology
- ○ **C.** Star topology
- ○ **D.** Ring topology

Q20 (2 marks)

A user accesses their email from multiple devices and wants emails to stay on the server so they can read them from any device.
Which email protocol should they use?

- ○ **A.** SMTP
- ○ **B.** POP3
- ○ **C.** IMAP
- ○ **D.** FTP

Q21 (2 marks)

A wireless network signal becomes weaker as users move further from the router.
Which characteristic of wireless connectivity does this demonstrate?

- ○ **A.** Bandwidth decreases with distance
- ○ **B.** Latency improves with distance
- ○ **C.** Security increases with distance
- ○ **D.** Range is limited and signal strength decreases with distance

Q22 (2 marks)

When data is transmitted across a network, one layer ensures that all packets arrive correctly and in the right order.
Which layer of the TCP/IP model handles this?

- ○ **A.** Application layer

- B. Transport layer
- C. Internet layer
- D. Link layer

Q23 (2 marks)

A video file is 1.8 gigabytes in size. The network connection speed is 120 Mbps.
How long will it take to download this file?
(1 byte = 8 bits, 1 gigabyte = 1000 megabytes)

- A. 15 seconds
- B. 2 minutes
- C. 15 minutes
- D. 20 minutes

Q24 (3 marks)

A student makes the following statements about TCP/IP:
Statement 1: TCP breaks data into packets and ensures reliable delivery
Statement 2: IP handles the addressing and routing of packets
Statement 3: TCP/IP is only used on local area networks, not the internet
Which statements are correct?

- A. Statement 1 only
- B. Statements 1 and 2 only
- C. Statements 2 and 3 only

- ○ **D.** All three statements are correct

Q25 (2 marks)

A school uses a star topology for its computer network.
What happens if the central switch fails?

- ○ **A.** Only the computers closest to the switch lose connection
- ○ **B.** All computers lose their network connection
- ○ **C.** Computers can still communicate directly with each other
- ○ **D.** The network automatically switches to a bus topology

Q26 (2 marks)

HTTP, HTTPS, FTP, and SMTP all operate at the same layer of the TCP/IP model.
Which layer do these protocols belong to?

- ○ **A.** Link layer
- ○ **B.** Internet layer
- ○ **C.** Transport layer
- ○ **D.** Application layer

Q27 (3 marks)

A company is deciding between wired and wireless networking for a new office.
Consider the following factors:

• Speed and reliability
• Installation cost
• Flexibility for moving devices
• Security

Which statement correctly compares wired and wireless options?

- ○ **A.** Wireless is better for all four factors because technology has improved significantly
- ○ **B.** Wired is better for speed, reliability, and security; wireless is better for flexibility and often cheaper to install
- ○ **C.** Wired and wireless perform identically across all four factors
- ○ **D.** Wireless is faster but less secure; wired is slower but more flexible

Q28 (3 marks)

A business needs to transfer 50 files. Each file is 400 megabytes. The network speed is 800 Mbps.
Calculate the total time needed to transfer all files.
(1 byte = 8 bits)

- ○ **A.** 200 seconds
- ○ **B.** 25 seconds
- ○ **C.** 2000 seconds
- ○ **D.** 3 minutes 20 seconds

Q29 (3 marks)

When a user sends an email, data passes through all four layers of the TCP/IP model.
Starting from the Application layer, what is the correct order of layers the data passes through before transmission?

- ○ **A.** Application → Internet → Transport → Link
- ○ **B.** Application → Transport → Internet → Link
- ○ **C.** Application → Link → Internet → Transport
- ○ **D.** Application → Transport → Link → Internet

Q30 (3 marks)

A network designer compares three topologies for a new network:
• Bus: Low cost, single point of failure
• Star: Moderate cost, central point of failure
• Mesh: High cost, highly resilient
For a hospital where network failure could affect patient safety, which topology is most appropriate and why?

- ○ **A.** Bus topology because it is the cheapest option for a large building
- ○ **B.** Star topology because it is easy to add new devices
- ○ **C.** Mesh topology because it provides multiple paths for data if one connection fails
- ○ **D.** Bus topology because a single cable is easier to maintain

Total: **56** marks • Time: **30 min**

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